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Course: CSE LAB

Course Code: 1202

Lab task Topic: Structured Programming Problems Solve (Loop)

Submitted to:

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Problem No. 01

Name of the problem:

Write a program in C to display n terms of natural number and their sum.

Test Data : 7

Expected Output :

The first 7 natural number is : 1 2 3 4 5 6 7

The Sum of Natural Number upto 7 terms : 28

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Displaying the numbers 1 to 10 using for loop
#include <stdio.h>
int main()
{
  int i,n,sum=0;
  printf("Input Value of terms : ");
  scanf("%d",&n);
  printf("\nThe first %d natural numbers are:\n",n);
  for(i=1;i<=n;i++)
  {
    printf("%d ",i);
    sum+=i;
  }
  printf("\nThe Sum of natural numbers upto %d terms : %d \n",n,sum);
}
```

Output:

```
"E:\Programming\my files\C++\looping\loop 1.exe"
Input Value of terms : 5

The first 5 natural numbers are:
1 2 3 4 5
The Sum of natural numbers upto 5 terms : 15

Process returned 0 (0x0)   execution time : 4.5
32 s
Press any key to continue.
```

Problem No. 02

Name of the problem:

Write a program in C to read 10 numbers from keyboard and find their sum and average.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Display the average of the values loop
#include <stdio.h>
int main()
{
   int i,n,sum=0;
        float avg;
        printf("Input the 10 numbers : \n");
        for (i=1;i<=10;i++)
        {
           printf("Number-%d :",i);
                scanf("%d",&n);
                sum +=n;
        }
        avg=sum/10.0;
        printf("The sum of 10 no is : %d\nThe Average is : %f\n",sum,avg);
}
```

Output:

```
"E:\Programming\my files\C++\looping\loop 2.exe"
Input the 10 numbers :
Number-1 :1
Number-2 :2
Number-3 :3
Number-4 :4
Number-5 :5
Number-6 :6
Number-7 :7
Number-8 :8
Number-9 :9
Number-10 :10
The sum of 10 no is : 55
The Average is : 5.500000

Process returned 0 (0x0)   execution time : 10
```

Problem No. 03

Name of the problem:

Write a program in C to display the cube of the number upto given an integer.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//display the cube of the number up to given an integer
#include <stdio.h>
int main()
{
  int i,n,c=0;
            float avg;
            printf("Input number of terms : \n");
            scanf("%d",&n);
            for (i=1;i<=n;i++)
            {
             c=i*i*i;
             printf("Number is : %d and cube of the %d is :%d\n",i,i,c);
            }
   return 0;
}
```

Output:

```
"E:\Programming\my files\C++\looping\loop 3.exe"
Input number of terms :
5
Number is : 1 and cube of the 1 is :1
Number is : 2 and cube of the 2 is :8
Number is : 3 and cube of the 3 is :27
Number is : 4 and cube of the 4 is :64
Number is : 5 and cube of the 5 is :125

Process returned 0 (0x0)   execution time : 3.
464 s
Press any key to continue.
```

Problem No. 04

Name of the problem:

Write a program in C to display the multiplication table of a given integer.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Table of inputed number
#include<stdio.h>
int main()
{
   int n,i,t=0;
   printf("Enter your n:");
   scanf("%d",&n);
   for(i=1;i<=10;i++)
   {
      t=n*i;
      printf("%d * %d = %d\n",n,i,t);
   }
   return 0;
}
```

Output:

```
"E:\Programming\my files\C++\looping\lab_loop_4.exe"
Enter your n:3
3 * 1 = 3
3 * 2 = 6
3 * 3 = 9
3 * 4 = 12
3 * 5 = 15
3 * 6 = 18
3 * 7 = 21
3 * 8 = 24
3 * 9 = 27
3 * 10 = 30
```

Problem No. 05

Name of the problem:

Write a program in C to display the multiplication table vertically from 1 to n.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Table of nth inputed number
#include<stdio.h>
int main()
{
   int n,i,t=0,x;
   printf("Enter your n:");
   scanf("%d",&n);

   for(i=1;i<=n;i++)
   {
      printf("\ntable of %d\n",i);
      for(x=1;x<=10;x++)
      {
         t=x*i;
         printf("%d * %d = %d\n",x,i,t);
      }
   }
   return 0;
}
```

Output

```
"E:\Programming\my files\C++\looping\lab_loop_4.exe"
Enter your n:3
3 * 1 = 3
3 * 2 = 6
3 * 3 = 9
3 * 4 = 12
3 * 5 = 15
3 * 6 = 18
3 * 7 = 21
3 * 8 = 24
3 * 9 = 27
3 * 10 = 30
```

Problem No. 06

Name of the problem:

Write a program in C to display the n terms of odd natural number and their sum

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//display the n terms of odd natural number and their sum
#include <stdio.h>
int main()
{
  int i,n,sum=0;


  printf("Input number of terms : ");
  scanf("%d",&n);
  printf("\nThe odd numbers are :");
  for(i=1;i<=n;i++)
  {
    printf("%d ",2*i-1);
    sum+=2*i-1;
  }
  printf("\nThe Sum of odd Natural Number upto %d terms : %d \n",n,sum);
}
```

Output:

```
"E:\Programming\my files\C++\looping\loop 6.exe"
Input number of terms : 10

The odd numbers are :1 3 5 7 9 11 13 15 17 19
The Sum of odd Natural Number upto 10 terms : 100

Process returned 0 (0x0)   execution time : 1.992 s
Press any key to continue.
```

Problem No. 07

Name of the problem:

Write a program in C to display the pattern like right angle triangle using an asterisk.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//* star looping triangle *-*
#include <stdio.h>
int main()
{
   int i,j;
   int n;
   printf("Enter triangle line numbers:");
   scanf("%d",&n);

   for(i=1; i<=n; i++)
   {
      for(j=1; j<=i; j++)
      printf("*");
      printf("\n");
   }
   return 0;
}
```

Output:

```
"E:\Programming\my files\C++\looping\lab_loop_7.exe"
Enter triangle line numbers:7
*
**
***
****
*****
******
*******

Process returned 0 (0x0)   execution time : 3.
660 s
Press any key to continue.
```

Problem No. 08

Name of the problem:

Write a program in C to display the pattern like right angle triangle with a number

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//display the pattern like right angle triangle with a number
#include <stdio.h>
int main()
{
  int i,j,rows;
  printf("Input number of rows : ");
  scanf("%d",&rows);
  for(i=1;i<=rows;i++)
  {
            for(j=1;j<=i;j++)
              printf("%d",j);
            printf("\n");
  }
}
```

Output:

```
"E:\Programming\my files\C++\looping\loop 8.exe"
Input number of rows : 8
1
12
123
1234
12345
123456
1234567
12345678

Process returned 0 (0x0)   execution time : 1.
995 s
```

Problem No. 09

Name of the problem:

Write a program in C to make such a pattern like right angle triangle with a number which will repeat a number in a row

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//pattern like right angle triangle with a number which will repeat a number in a row.
#include <stdio.h>
void main()
{
  int i,j,rows;


  printf("Input number of rows : ");
  scanf("%d",&rows);
  for(i=1;i<=rows;i++)
  {
            for(j=1;j<=i;j++)
              printf("%d",i);
            printf("\n");
  }
}
```

Output:

```
"E:\Programming\my files\C++\looping\loop 9.exe"
Input number of rows : 9
1
22
333
4444
55555
666666
7777777
88888888
999999999

Process returned 9 (0x9)   execution time : 0.
```

Problem No. 10

Name of the problem:

Write a program in C to make such a pattern like right angle triangle with number increased by 1.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//pattern like right angle triangle with a number which will repeat a number in a row.
#include <stdio.h>
void main()
{
  int i,j,rows;


  printf("Input number of rows : ");
  scanf("%d",&rows);
  for(i=1;i<=rows;i++)
  {
            for(j=1;j<=i;j++)
              printf("%d",i);
            printf("\n");
  }
}
```

Output:

```
"E:\Programming\my files\C++\looping\loop 10.exe"
Input number of rows : 6
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
16 17 18 19 20 21

Process returned 0 (0x0)   execution time : 5.
773 s
Press any key to continue.
```

Problem No. 11

Name of the problem:

Write a program in C to make such a pattern like a pyramid with numbers increased by 1.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//pattern like a pyramid with numbers increased by 1.
#include <stdio.h>
int main()
{
  int i,j,spc,rows,k,t=1;
  printf("Input number of rows : ");
  scanf("%d",&rows);
  spc=rows+4-1;
  for(i=1;i<=rows;i++)
  {
      for(k=spc;k>=1;k--)
        {
          printf(" ");
        }
                for(j=1;j<=i;j++)
                printf("%d ",t++);
              printf("\n");
   spc--;
  }}
```

Output:

```
"E:\Programming\my files\C++\looping\loop 11.exe"
Input number of rows : 4
        1
       2 3
      4 5 6
     7 8 9 10

Process returned 0 (0x0)   execution time : 0.
708 s
Press any key to continue.
```

THE END